Лабораторная работа №7

Дисциплина: Сетевые технологии

Пакавира Арсениу Висенте Луиш

2026-03-20

1 Цель работы

- Настройка службы DHCP
- Распределение адресов IPv4 и IPv6
- Анализ DHCP-трафика

2 Топология и захват трафика (IPv4 DHCP)

- Переименование устройств по шаблону
- PC1-bahis, bahis-sw-01, bahis-gw-01
- Включён захват трафика между коммутатором и маршрутизатором

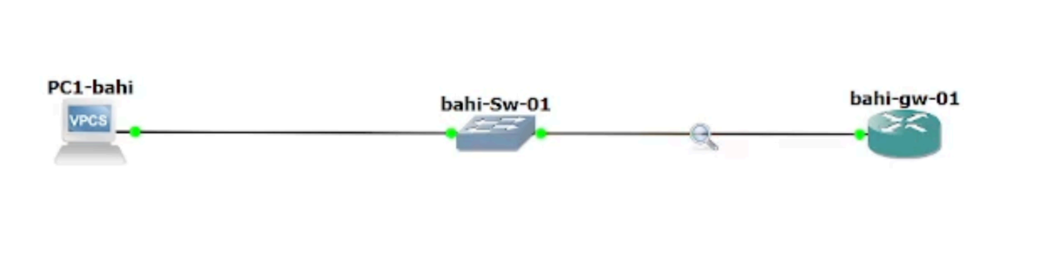
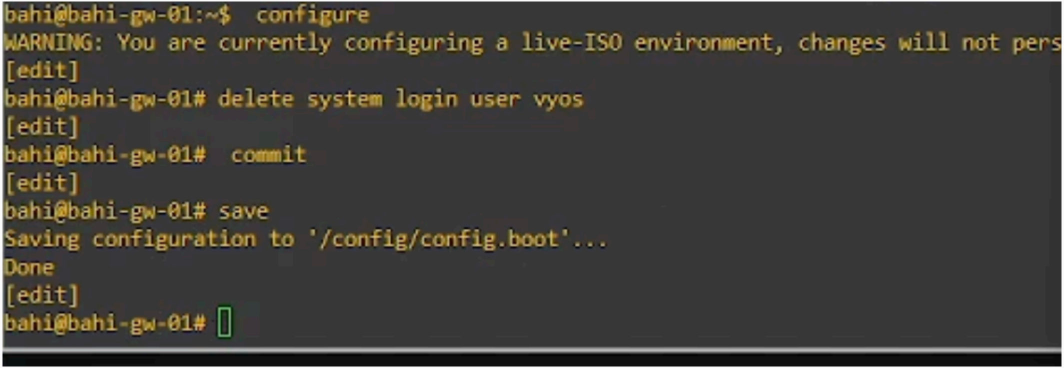


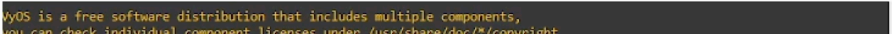
Рисунок 1: Топология сети в GNS3 с переименованными устройствами и включённым захватом трафика

3 Системные параметры VyOS и пользователи

- Задано имя маршрутизатора
- Настроено доменное имя
- Создан пользователь bahis
- Удалён пользователь vyos



```
bahi@bahi-gw-01:~$  configure
WARNING: You are currently configuring a live-ISO environment, changes will not pers
[edit]
bahi@bahi-gw-01# delete system login user vyos
[edit]
bahi@bahi-gw-01#  commit
[edit]
bahi@bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
bahi@bahi-gw-01#
```



```
VyOS is a free software distribution that includes multiple components,
```

4 Настройка DHCP-сервера (IPv4)

- Создан shared-network bahis
- DNS: `10.0.0.1`
- Gateway: `10.0.0.1`
- Диапазон: `10.0.0.2–10.0.0.253`

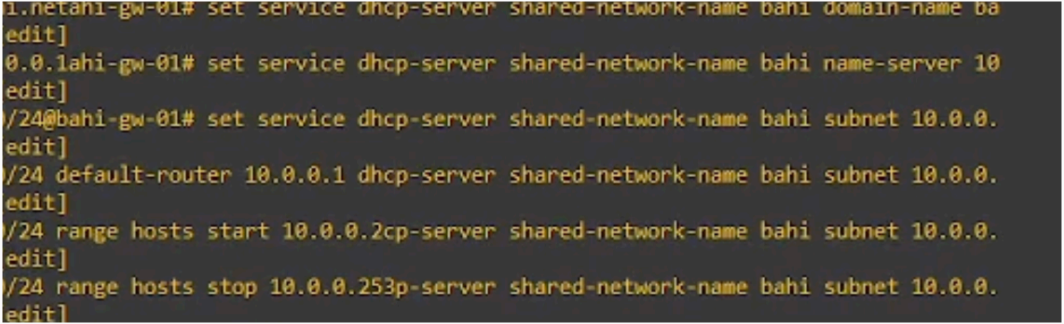


Рисунок 2: Конфигурация DHCP-сервера на маршрутизаторе VyOS для сети 10.0.0.0/24

5 Статистика DHCP-сервера (до клиентов)

- Всего адресов: 252
- Активных аренд: 0
- Все адреса доступны

```
bahi@bahi-gw-01:~$ show dhcp server statistics
Pool       Size    Leases    Available   Usage
------   ------   --------   ----------- -------
bahi        252          0          252  0%
bahi@bahi-gw-01:~$ show dhcp server leases
IP address    Hardware address    State    Lease start    Lease expiration    Remaining    Pool    H
name
-----------  -----------------  -------  ------------  -----------------  ----------  ------  -
------
bahi@bahi-gw-01:~$
```

Рисунок 3: Статистика DHCP-сервера и отсутствие активных аренд IP-адресов

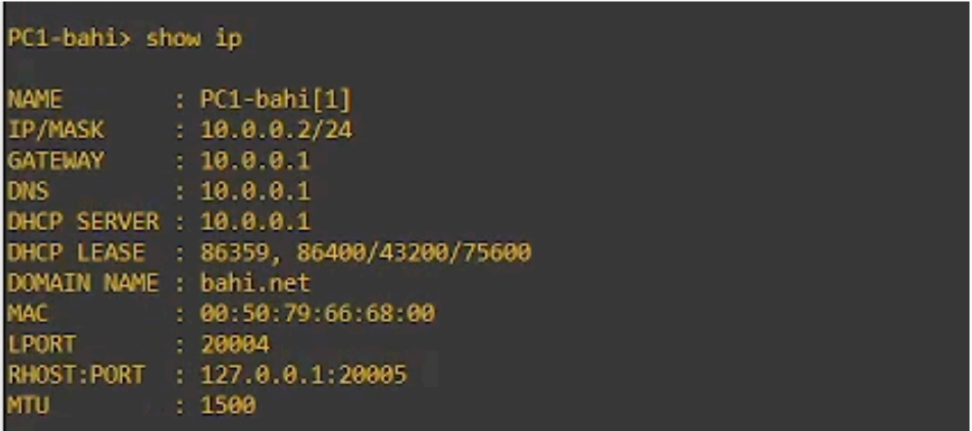
6 DHCP-процесс на узле PC1 (Request / ACK)

- DHCP Request
- DHCP ACK
- Назначен IP `10.0.0.2/24`

```
Option 12: Host Name = PC1-bahi
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = bahi.net

Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00
Option 12: Host Name = PC1-bahi

Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
```

7 Проверка IP и связности PC1

- IP получен по DHCP
- Gateway: `10.0.0.1`
- Ping маршрутизатора успешен



```
PC1-bahi> show ip

NAME        : PC1-bahi[1]
IP/MASK     : 10.0.0.2/24
GATEWAY     : 10.0.0.1
DNS         : 10.0.0.1
DHCP SERVER : 10.0.0.1
DHCP LEASE  : 86359, 86400/43200/75600
DOMAIN NAME : bahi.net
MAC         : 00:50:79:66:68:00
LPORT       : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500
```

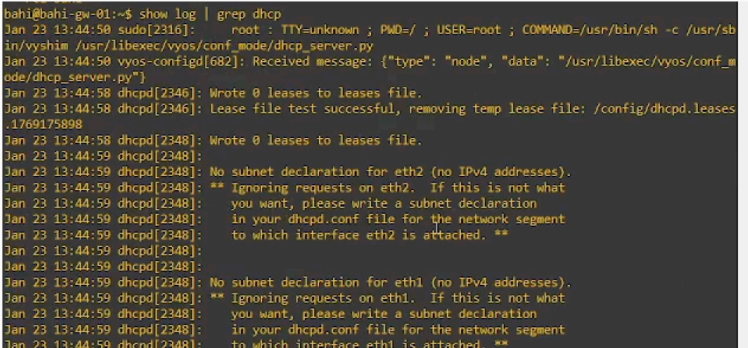
8 DHCP-статистика после выдачи адреса

- Активных аренд: 1
- IP: `10.0.0.2`
- Клиент: PC1-bahis

```
-vbash: /config/dhcpd.leases: Permission denied
bahi@bahi-gw-01:~$ show dhcp server statistics
Pool      Size    Leases    Available  Usage
------  ------  --------  -----------  -------
bahi       252         0          252  0%
bahi@bahi-gw-01:~$ show dhcp server leases
IP address    Hardware address    State    Lease start    Lease expiration    Remaining    Pool    Host
name
------------  ------------------  -------  -------------  ------------------  -----------  ------  ----
------
bahi@bahi-gw-01:~$ show dhcp server statistics
Pool      Size    Leases    Available  Usage
------  ------  --------  -----------  -------
bahi       252         1          251  0%
bahi@bahi-gw-01:~$ show dhcp server leases
IP address    Hardware address    State    Lease start         Lease expiration      Remaining     Pool
   Hostname
```

9 Журнал DHCP-сервера

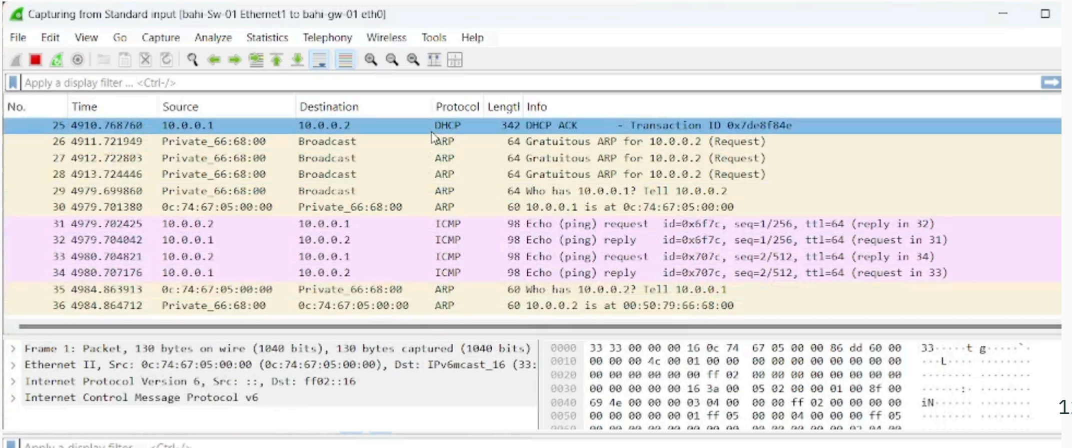
- DHCPDISCOVER
- DHCPOFFER
- DHCPREQUEST
- DHCPACK
- Обслуживание клиента PC1



```
bahi@bahi-gw-01:~$ show log | grep dhcp
Jan 23 13:44:50 sudo[2316]:     root : TTY=unknown ; PWD=/ ; USER=root ; COMMAND=/usr/bin/sh -c /usr/sb
in/vyshim /usr/libexec/vyos/conf_mode/dhcp_server.py
Jan 23 13:44:50 vyos-configd[682]: Received message: {"type": "node", "data": "/usr/libexec/vyos/conf_m
ode/dhcp_server.py"}
Jan 23 13:44:58 dhcpd[2346]: Wrote 0 leases to leases file.
Jan 23 13:44:58 dhcpd[2346]: Lease file test successful, removing temp lease file: /config/dhcpd.leases
.1769175898
Jan 23 13:44:58 dhcpd[2348]: Wrote 0 leases to leases file.
Jan 23 13:44:59 dhcpd[2348]:
Jan 23 13:44:59 dhcpd[2348]: No subnet declaration for eth2 (no IPv4 addresses).
Jan 23 13:44:59 dhcpd[2348]: ** Ignoring requests on eth2.  If this is not what
Jan 23 13:44:59 dhcpd[2348]:    you want, please write a subnet declaration
Jan 23 13:44:59 dhcpd[2348]:    in your dhcpd.conf file for the network segment
Jan 23 13:44:59 dhcpd[2348]:    to which interface eth2 is attached. **
Jan 23 13:44:59 dhcpd[2348]:
Jan 23 13:44:59 dhcpd[2348]:
Jan 23 13:44:59 dhcpd[2348]: No subnet declaration for eth1 (no IPv4 addresses).
Jan 23 13:44:59 dhcpd[2348]: ** Ignoring requests on eth1.  If this is not what
Jan 23 13:44:59 dhcpd[2348]:    you want, please write a subnet declaration
Jan 23 13:44:59 dhcpd[2348]:    in your dhcpd.conf file for the network segment
Jan 23 13:44:59 dhcpd[2348]:    to which interface eth1 is attached. **
```

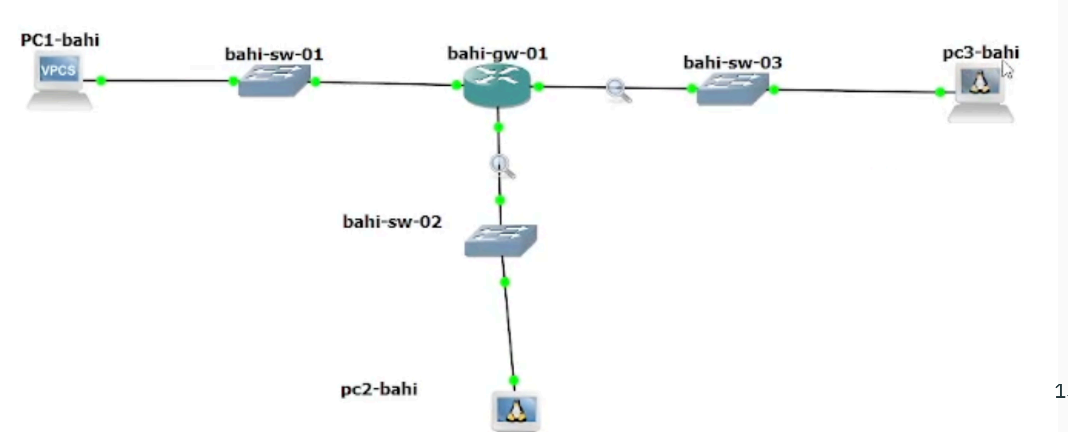
10 Анализ DHCP-трафика (IPv4)

- Discover → Offer → Request → ACK
- ARP и Gratuitous ARP
- ICMP Echo



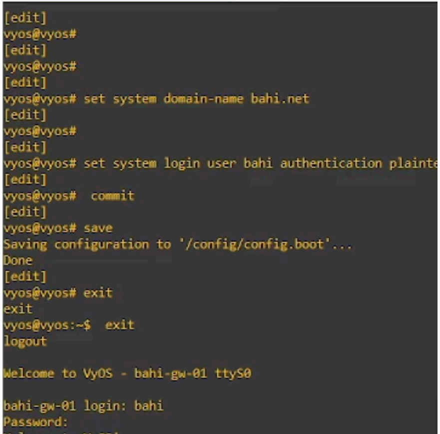
11 IPv6: топология и захват трафика

- Использован Alpine Linux
- Захват ICMPv6 и DHCPv6
- Подсети IPv6



12 Системные параметры VyOS (IPv6)

- Имя устройства
- Доменное имя
- Пользователь bahis



13 Назначение IPv6-адресов интерфейсам

- eth1: `2000::1/64`
- eth2: `2001::1/64`

```
bahi@bahi-gw-01:~$ configure
WARNING: You are currently configuring a live-ISO environment, changes will not persist until installed
[edit]
bahi@bahi-gw-01# delete system login user vyos
[edit]
bahi@bahi-gw-01# commit
[edit]
bahi@bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
bahi@bahi-gw-01# set interfaces ethernet eth1 address 2000::1/64
[edit]
bahi@bahi-gw-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
bahi@bahi-gw-01# show interfaces
 ethernet eth0 {
     hw-id 0c:fd:55:1c:00:00
 }
 ethernet eth1 {
+    address 2000::1/64
     hw-id 0c:fd:55:1c:00:01
 }
 ethernet eth2 {
+    address 2001::1/64
     hw-id 0c:fd:55:1c:00:02
```

14 Stateless DHCPv6 + Router Advertisement

- SLAAC
- other-config-flag
- DNS и домен по DHCPv6

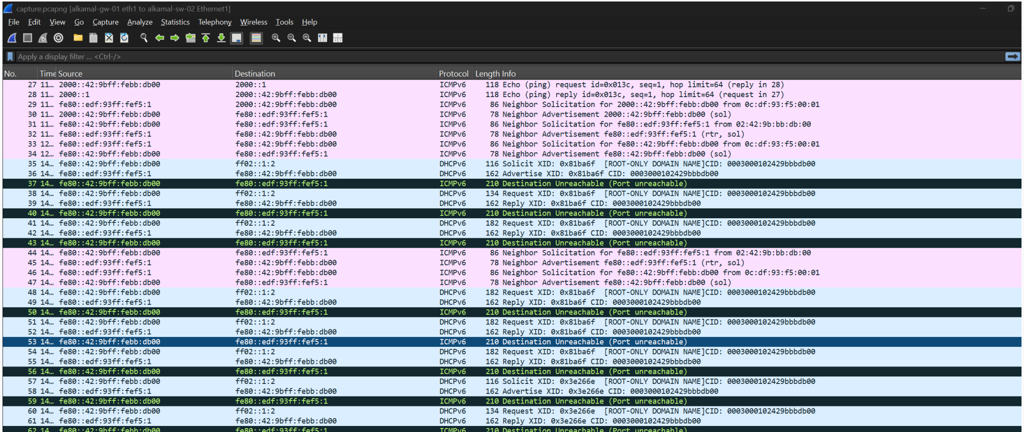
```
bahi@bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
bahi@bahi-gw-01# set service router-advert interface eth1 prefix 2000::/64
[edit]
bahi@bahi-gw-01# set service router-advert interface eth1 other-config-flag
[edit]
bahi@bahi-gw-01# set service dhcpv6-server shared-network-name bahi-stateless
[edit]
bnet 2000::0/64# set service dhcpv6-server shared-network-name bahi-stateless su
[edit]
mmon-options name-server 2000::1pv6-server shared-network-name bahi-stateless co

[edit]
bahi@bahi-gw-01#
[edit]
mmon-options domain-search bahi.net-server shared-network-name bahi-stateless co
[edit]
```

15 Проверка IPv6 на PC2 (Stateless)

- SLAAC-адрес
- Link-local адрес
- Ping успешен

```
pc2-bahi console is now available... Press RETURN to get started.
/ # ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
6: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN qlen 1000
    link/ether 02:42:e8:69:53:00 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::42:e8ff:fe69:5300/64 scope link
       valid_lft forever preferred_lft forever
/ # ip -6 route show
fe80::/64 dev eth0  metric 256
/ # /# ping 2000::1 -c 2
/bin/sh: /#: not found
/ #  ping 2000::1 -c 2
PING 2000::1 (2000::1): 56 data bytes
ping: sendto: Network unreachable
/ # cat /etc/resolv.conf
/ # udhcpc6 -i eth0
udhcpc6: started, v1.37.0
udhcpc6: sending discover
udhcpc6: sending discover
udhcpc6: sending discover
udhcpc6: sending discover
```

- Router Solicitation
- Router Advertisement
- DHCPv6 Solicit / Reply



17 Настройка Stateful DHCPv6

- managed-flag
- Диапазон: `2001::100–2001::199`
- DNS и домен

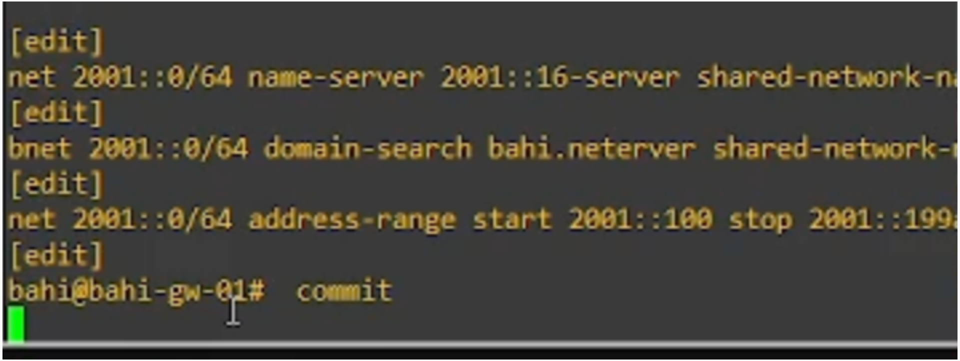


Рисунок 12: Конфигурация Stateful DHCPv6

18 Получение IPv6-адреса PC3

- До запроса: только link-local
- После udhcpc6: глобальный IPv6

```
PC3-alkamal console is now available... Press RETURN to get started.
/ #  ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
10: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN qlen 1000
    link/ether 02:42:88:32:29:00 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::42:88ff:fe32:2900/64 scope link
       valid_lft forever preferred_lft forever
/ # ip -6 route show
fe80::/64 dev eth0  metric 256
/ #  cat /etc/resolv.conf
/ #  udhcpc6 -i eth0
udhcpc6: started, v1.37.0
udhcpc6: sending discover
udhcpc6: sending select
udhcpc6: IPv6 obtained, lease time 43200
```


Рисунок 13: Получение IPv6 по DHCPv6

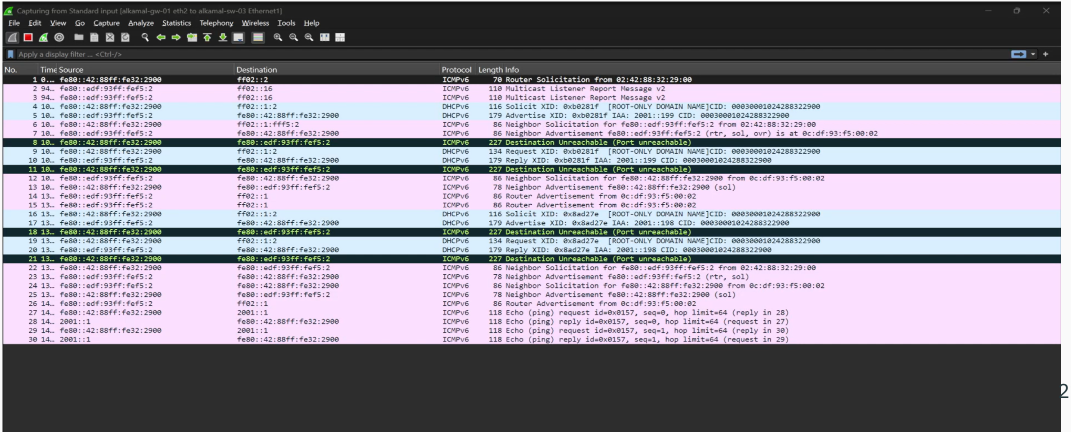
19 Проверка IPv6 PC3 после DHCPv6

- Глобальный IPv6
- Default route
- DNS получен

```
/ #  ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
10: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN qlen 1000
    link/ether 02:42:88:32:29:00 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::42:88ff:fe32:2900/64 scope link
       valid_lft forever preferred_lft forever
/ # ip -6 route show
fe80::/64 dev eth0  metric 256
/ # ping 2001::1 -c 2
PING 2001::1 (2001::1): 56 data bytes
ping: sendto: Network unreachable
/ #  cat /etc/resolv.conf
search alkamal.net
nameserver 2001:0000:0000:0000:0000:0000:0000:0001
/ #
```

20 Анализ DHCPv6 Stateful-трафика

- RA с managed-flag
- Solicit → Advertise → Request → Reply
- Адрес успешно выдан



Capturing from Standard input [alkamal-gw-01 eth2 to alkamal-sw-03 Ethernet1]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter … <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0…	fe80::42:88ff:fe32:2900	ff02::2	ICMPv6	70	Router Solicitation from 02:42:88:32:29:00
2	94…	fe80::edf:93ff:fef5:2	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
3	94…	fe80::edf:93ff:fef5:2	ff02::16	ICMPv6	110	Multicast Listener Report Message v2
4	10…	fe80::42:88ff:fe32:2900	ff02::1:2	DHCPv6	116	Solicit XID: 0xb0281f [ROOT-ONLY DOMAIN NAME]CID: 00030001024288322900
5	10…	fe80::edf:93ff:fef5:2	fe80::42:88ff:fe32:2900	DHCPv6	179	Advertise XID: 0xb0281f IAA: 2001::199 CID: 00030001024288322900
6	10…	fe80::42:88ff:fe32:2900	ff02::1:fff5:2	ICMPv6	86	Neighbor Solicitation for fe80::edf:93ff:fef5:2 from 02:42:88:32:29:00
7	10…	fe80::edf:93ff:fef5:2	fe80::42:88ff:fe32:2900	ICMPv6	86	Neighbor Advertisement fe80::edf:93ff:fef5:2 (rtr, sol, ovr) is at 0c:df:93:f5:00:02
8	10…	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fef5:2	ICMPv6	227	Destination Unreachable (Port unreachable)
9	10…	fe80::42:88ff:fe32:2900	ff02::1:2	DHCPv6	134	Request XID: 0xb0281f [ROOT-ONLY DOMAIN NAME]CID: 00030001024288322900
10	10…	fe80::edf:93ff:fef5:2	fe80::42:88ff:fe32:2900	DHCPv6	179	Reply XID: 0xb0281f IAA: 2001::199 CID: 00030001024288322900
11	10…	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fef5:2	ICMPv6	227	Destination Unreachable (Port unreachable)
12	10…	fe80::edf:93ff:fef5:2	fe80::42:88ff:fe32:2900	ICMPv6	86	Neighbor Solicitation for fe80::42:88ff:fe32:2900 from 0c:df:93:f5:00:02
13	10…	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fef5:2	ICMPv6	78	Neighbor Advertisement fe80::42:88ff:fe32:2900 (sol)
14	13…	fe80::edf:93ff:fef5:2	ff02::1	ICMPv6	86	Router Advertisement from 0c:df:93:f5:00:02
15	13…	fe80::edf:93ff:fef5:2	ff02::1	ICMPv6	86	Router Advertisement from 0c:df:93:f5:00:02
16	13…	fe80::42:88ff:fe32:2900	ff02::1:2	DHCPv6	116	Solicit XID: 0x8ad27e [ROOT-ONLY DOMAIN NAME]CID: 00030001024288322900
17	13…	fe80::edf:93ff:fef5:2	fe80::42:88ff:fe32:2900	DHCPv6	179	Advertise XID: 0x8ad27e IAA: 2001::198 CID: 00030001024288322900
18	13…	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fef5:2	ICMPv6	227	Destination Unreachable (Port unreachable)
19	13…	fe80::42:88ff:fe32:2900	ff02::1:2	DHCPv6	134	Request XID: 0x8ad27e [ROOT-ONLY DOMAIN NAME]CID: 00030001024288322900
20	13…	fe80::edf:93ff:fef5:2	fe80::42:88ff:fe32:2900	DHCPv6	179	Reply XID: 0x8ad27e IAA: 2001::198 CID: 00030001024288322900
21	13…	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fef5:2	ICMPv6	227	Destination Unreachable (Port unreachable)
22	13…	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fef5:2	ICMPv6	86	Neighbor Solicitation for fe80::edf:93ff:fef5:2 from 02:42:88:32:29:00
23	13…	fe80::edf:93ff:fef5:2	fe80::42:88ff:fe32:2900	ICMPv6	78	Neighbor Advertisement fe80::edf:93ff:fef5:2 (rtr, sol)
24	13…	fe80::edf:93ff:fef5:2	fe80::42:88ff:fe32:2900	ICMPv6	86	Neighbor Solicitation for fe80::42:88ff:fe32:2900 from 0c:df:93:f5:00:02
25	13…	fe80::42:88ff:fe32:2900	fe80::edf:93ff:fef5:2	ICMPv6	78	Neighbor Advertisement fe80::42:88ff:fe32:2900 (sol)
26	14…	fe80::edf:93ff:fef5:2	ff02::1	ICMPv6	86	Router Advertisement from 0c:df:93:f5:00:02
27	14…	fe80::42:88ff:fe32:2900	2001::1	ICMPv6	118	Echo (ping) request id=0x0157, seq=0, hop limit=64 (reply in 28)
28	14…	2001::1	fe80::42:88ff:fe32:2900	ICMPv6	118	Echo (ping) reply id=0x0157, seq=0, hop limit=64 (request in 27)
29	14…	fe80::42:88ff:fe32:2900	2001::1	ICMPv6	118	Echo (ping) request id=0x0157, seq=1, hop limit=64 (reply in 30)
30	14…	2001::1	fe80::42:88ff:fe32:2900	ICMPv6	118	Echo (ping) reply id=0x0157, seq=1, hop limit=64 (request in 29)